

Consider the problem of scheduling the assembly of a car. We consider a small part of the car assembly, consisting of 15 tasks: install axles (front and back), affix all four wheels (right and left, front and back), tighten nuts for each wheel, affix hubcaps for each wheel, and inspect the final assembly.

The axles have to be in place before the wheels are put on, and it takes 10 minutes to install an axle. Next for each wheel, we must affix the wheel, then tighten the nuts, and finally attach the hubcap. It takes 1 minute to affix each wheel, 2 minutes to tighten these nuts for each wheel, and 1 minute to attach each hubcap.

Suppose we have four workers to install wheels, but they have to share one tool that helps put the axle in place. So the two axles must be placed in separate times. We also need to assert that the inspection comes last and takes 3 minutes. Also, note that each task starts at a discrete integer time point and the whole assembly must be done in 30 minutes.

1. Formulate the problem as a Constraint Satisfaction Problem (CSP) by identifying the variables, their domains, and the associated constraints using formal notation.

11
(CO2)
(PO2)

Solution:

We represent the tasks with 15 variables:

$$Axle_F, Axle_B, Wheel_{RF}, Wheel_{LF}, Wheel_{RB}, Wheel_{LB}, Nuts_{RF}, \\ Nuts_{LF}, Nuts_{RB}, Nuts_{LB}, Cap_{RF}, Cap_{LF}, Cap_{RB}, Cap_{LB}, Inspect$$

The domain for each variable will be: $\{0, 1, 2, \dots, 30\}$.

We apply precedence constraints between individual tasks. For the axles, we write:

$$Axle_F + 10 \leq Wheel_{RF}; \quad Axle_F + 10 \leq Wheel_{LF}; \\ Axle_B + 10 \leq Wheel_{RB}; \quad Axle_B + 10 \leq Wheel_{LB}$$

For wheels, nuts, and hubcaps, we can write:

$$Wheel_{RF} + 1 \leq Nuts_{RF}; \quad Nuts_{RF} + 2 \leq Cap_{RF}; \\ Wheel_{LF} + 1 \leq Nuts_{LF}; \quad Nuts_{LF} + 2 \leq Cap_{LF}; \\ Wheel_{RB} + 1 \leq Nuts_{RB}; \quad Nuts_{RB} + 2 \leq Cap_{RB}; \\ Wheel_{LB} + 1 \leq Nuts_{LB}; \quad Nuts_{LB} + 2 \leq Cap_{LB}.$$

For the placement of the two axles, we write:

$$(Axle_F + 10 \leq Axle_B) \text{ or } (Axle_B + 10 \leq Axle_F).$$

For the hubcap and inspection, we write:

$$Inspect + 3 \leq 30; \\ Cap_{RF} + 1 \leq Inspect; \quad Cap_{LF} + 1 \leq Inspect; \\ Cap_{RB} + 1 \leq Inspect; \quad Cap_{LB} + 1 \leq Inspect.$$

Rubric:

- 1 point for each type of variable
- 1 point for domain
- 1 point for each type of constraint

2. Assume that your formulation has a cutset of size $(X \bmod N)$, where X is the last digit of your student ID and N is the number of variables. You are running your CSP solver on a computer that can process 10^9 nodes/second.

2×2
(CO1)
(PO1)

Determine the expected time required in the worst-case to solve the CSP using the following algorithms:

- a) Naive backtracking

Solution:

$$\begin{aligned}\text{Time Required} &= \frac{d^N}{10^9} \\ &= \frac{30^{15}}{10^9} \text{ seconds}\end{aligned}$$

Rubric:

- 2 points for the correct equation

b) Nearly Tree-structured CSP

Solution:

$$\begin{aligned}\text{Time Required} &= \frac{d^c(N - c)d^2}{10^9} \\ &= \frac{30^X(15 - X)30^2}{10^9}\end{aligned}$$

Rubric:

- 2 points for the correct equation